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DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H1 BIOLOGY

8875/01

Paper 1 Multiple Choice Questions

30 September 2013

1 hour

Additional Materials: OTAS sheet

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section A MCQ [30 marks]

There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This document consists of **16** printed pages and **0** blank page.

Answer **all** questions in this section.

- 1 **Fig. 1** below is an electron micrograph of a human cell.

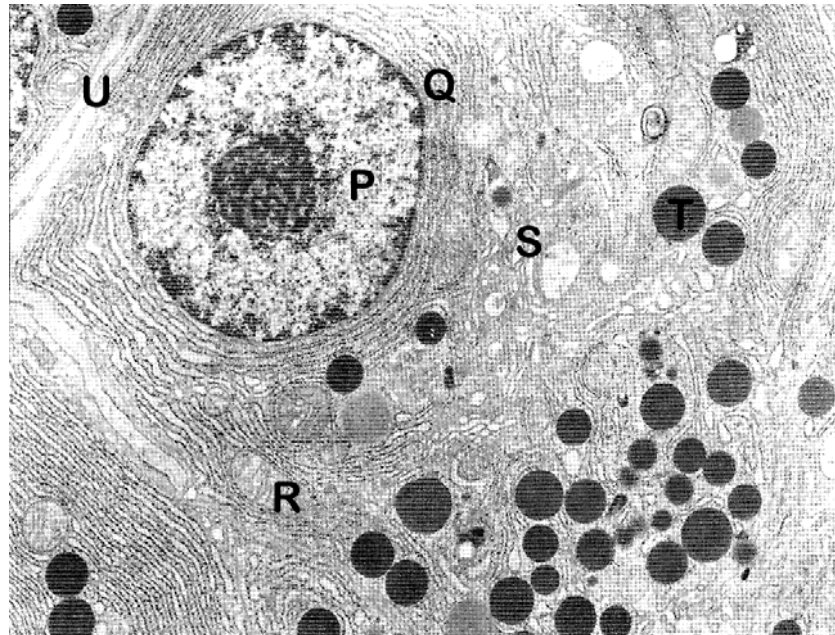


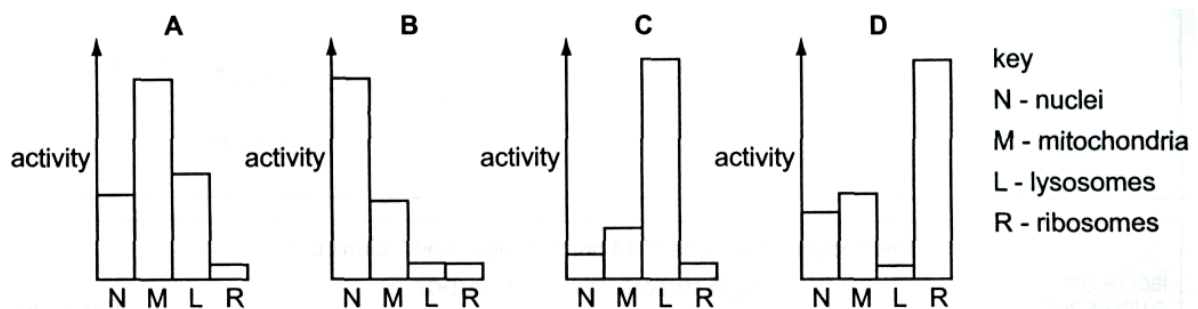
Fig. 1

Which of the following correctly identifies its structure-function adaptation?

<i>function</i>	<i>structural adaptation</i>
A antibody production	prominent P
B enzyme secretion	extensive Q , S and T
C phagocytosis	extensive R
D absorption	extensively-folded U

- 2 A piece of mammalian tissue was homogenised and subjected to differential centrifugation to yield four subcellular fractions. The activity within each fraction, of four different types of enzyme, **A**, **B**, **C** and **D**, was investigated.

Which bar chart shows the results of investigating hydrolytic enzyme activity?



3 Membranes in cells include the following components.

- 1 cholesterol
- 2 glycoproteins
- 3 phospholipids
- 4 proteins

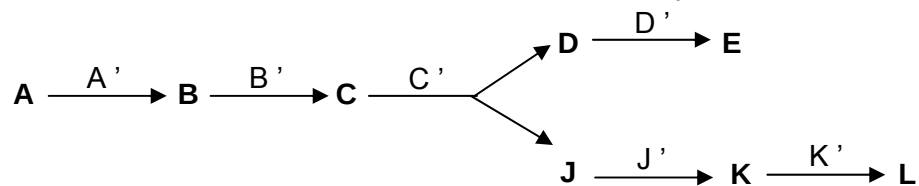
Which component is the most important for these functions of membranes?

function	recognising self/ non self	separating dissolved ions	stabilising the hydrophobic layer	transporting ions through membranes
A	1	2	3	4
B	2	3	1	4
C	3	1	4	2
D	3	4	1	2

4 In amylopectin, the bonds found between glucose units are _____.

- A α (1 $\rightarrow$ 4) glycosidic bonds only
- B α (1 $\rightarrow$ 4) glycosidic bonds and α (1 $\rightarrow$ 6) glycosidic bonds only
- C β (1 $\rightarrow$ 4) glycosidic bonds only
- D β (1 $\rightarrow$ 4) glycosidic bonds and β (1 $\rightarrow$ 6) glycosidic bonds only

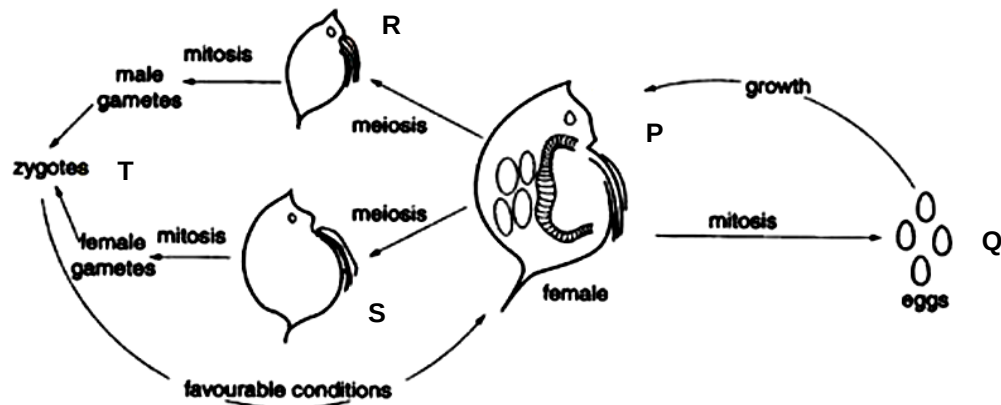
5 The figure shows a series of reactions in a metabolic pathway.



Enzyme **C'** catalyses the splitting of **C** into **D** and **J**. Assuming that product **E** is an allosteric inhibitor of **C'**, which of the following would likely happen if **E** were not consumed in a subsequent reaction?

- A The rate of production of **D** would increase.
- B The rate of production of **E** would remain the same.
- C The rate of production of **L** would remain the same.
- D The rate of production of all products **D**, **E**, **J**, **K** would decrease.

- 6 The figure shows the life cycle of the water flea, *Daphnia*. The cells of individual **R** contain 10 chromosomes.



Which of the following are correct?

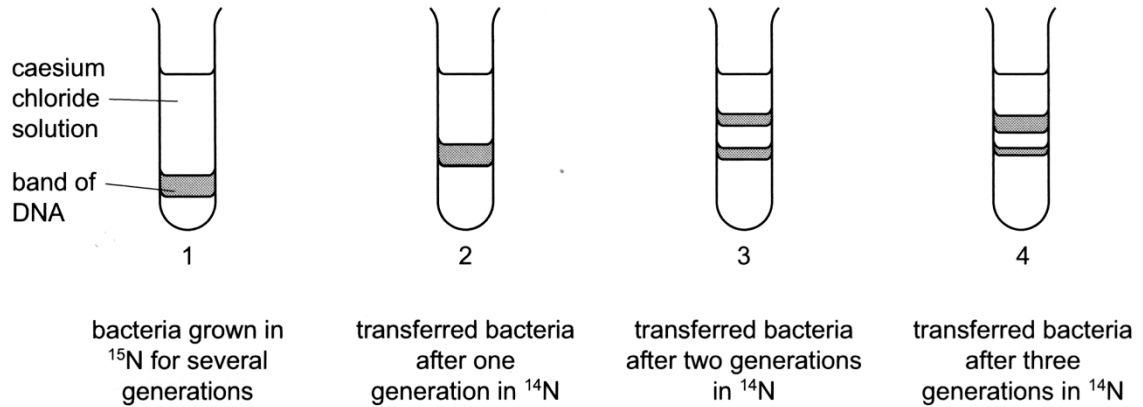
	Individual	Ploidy level	Number of chromosomes	Reason for choice
I	P	2n	20	The cells of P can undergo both mitosis and meiosis.
II	Q	2n	20	P produces eggs Q by mitosis which develop into females.
III	S	n	10	The gametic cells of P have undergone normal meiosis.
IV	T	2n	20	Random fertilisation of haploid gametes from R and S occurred to form zygote T .

- A I and II only
 B I and III only
 C II and IV only
 D All of the above
- 7 What is the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?
- I The anti-parallel arrangement of the DNA strands.
 II The RNA primers are required to initiate DNA elongation.
 III DNA polymerase joins new nucleotides to the 3' end of the growing strand.
 IV Helicase and single-stranded binding proteins work at the 5' end of the DNA strand.

- A I and III
 B II and IV
 C I and IV
 D II and III

- 8 Bacteria grown in ^{15}N for many generations were transferred to ^{14}N for further replication. DNA from the bacteria was extracted and separated by density gradient centrifugation.

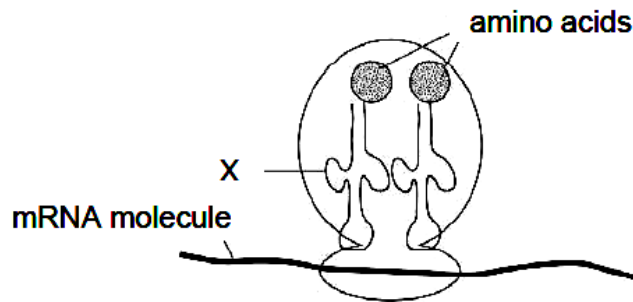
Their results are summarised in the following diagram.



What are the percentages of DNA containing ^{15}N and ^{14}N in tube 4?

	percentage DNA	
	^{15}N	^{14}N
A	0	100
B	10	90
C	12.5	87.5
D	15	85

- 9 The diagram below shows part of a molecule of mRNA bound to a ribosome.



Which of the following is **FALSE** about molecule X?

- I. It is formed by RNA polymerase in the nucleus.
- II. It is able to form hydrogen bonds with mRNA.
- III. An amino acid was attached to it by the enzyme amino-acyl tRNA transferase.
- IV. It is held in the amino-acyl tRNA binding site of the ribosome.

- A III only
- B IV only
- C I and II
- D III and IV

- 10 Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein it codes for?

	Promoter	Transcription terminator	Start codon	Stop codon	Middle of an intron
A	No protein product is produced	Protein product is shorter than normal	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
B	Too much protein product is produced	Protein product is normal	No protein product is produced	Protein product is longer than normal	Protein product is normal
C	Protein product is normal	Protein product is longer than normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Protein product is longer than normal	Too much protein product is produced	Protein product is normal	Protein product is shorter than normal	No protein product is produced

- 11** A cross between a round-leafed, tall plant and round-leafed dwarf plant produced the following offspring:

Key

121 round-leafed, tall plant

R – round leaf

124 round-leafed, dwarf plant

r – oval leaf

42 oval-leafed, tall plant

T – tall

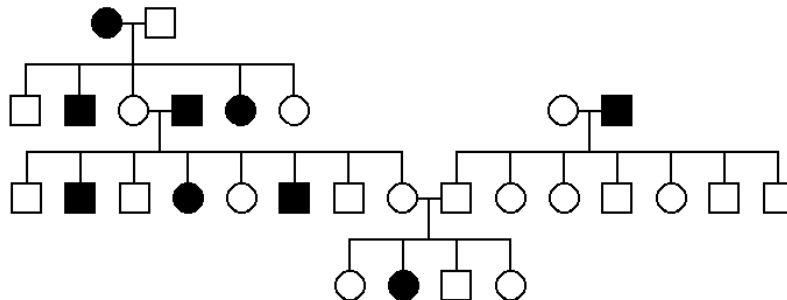
37 oval-leafed, dwarf plant

t – dwarf

What were the genotypes of the parents?

- A** **RrTt** x **Rrtt**
B **RrTt** x **RRtt**
C **RrTT** x **Rrtt**
D **RrTT** x **RRtt**

- 12** The inheritance of a disease in a family is presented in a pedigree tree below.



What type of inheritance could this disease show?

- | | |
|-----|----------------------|
| I | Autosomal dominant |
| II | Autosomal recessive |
| III | Sex-linked dominant |
| IV | Sex-linked recessive |

- A** I only
B II only
C I and III
D II and IV

- 13 In Shorthorn cattle, the allele for hornless is dominant to the allele for the presence of horns. Coat colour can be red (genotype $C^R C^R$), roan (genotype $C^R C^W$) or white (genotype $C^W C^W$).

A roan bull, heterozygous for the hornless trait, is crossed with a cow of the same genotype.

What is the probability that a calf from this cross would have the same genotype as its parents?

- A 1/4
- B 3/8
- C 1/2
- D 3/4

- 14
1. Light joins carbon dioxide to an acceptor compound which is then reduced by hydrogen obtained from water.
 2. Carbon dioxide combines with an acceptor compound and this is reduced by hydrogen split from water by light
 3. Light splits carbon dioxide and the resulting carbon then combines with oxygen and hydrogen obtained from water.
 4. Light splits water and the resulting hydroxyl group combines with a compound which has incorporated carbon dioxide

Which of the following statement is/are **CORRECT** outline of the main events in photosynthesis?

- A 1 only
- B 2 only
- C 1 and 2
- D 3 and 4

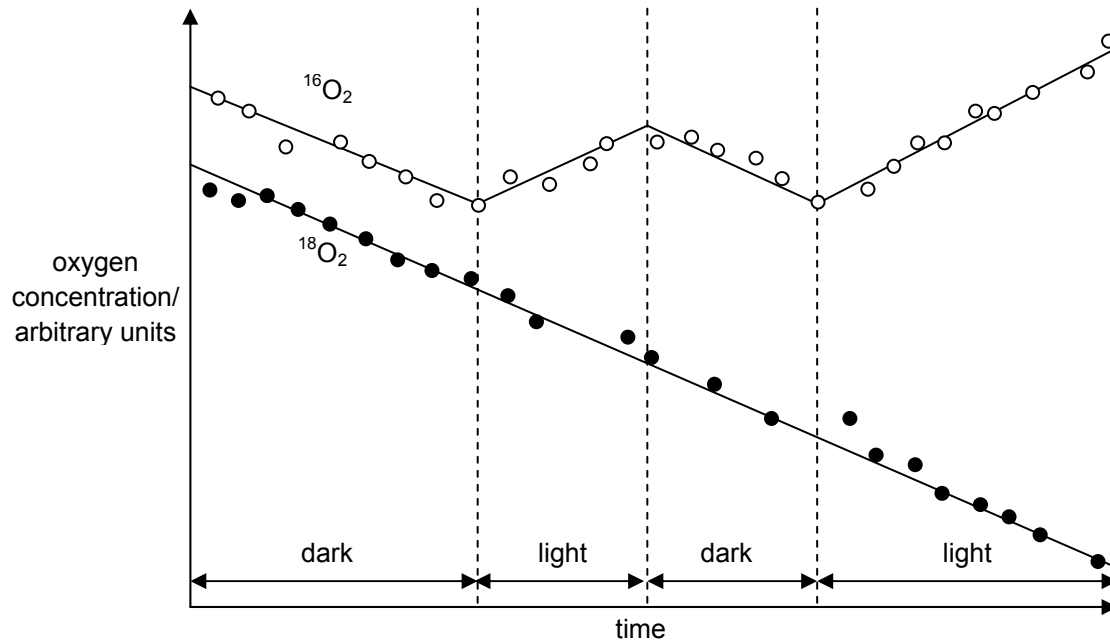
- 15 Dinitrophenol is a metabolic poison that can lodge within the thylakoid membranes of chloroplasts. It then provides an alternative route for H^+ ions to diffuse across the thylakoid membranes. In what way will the Calvin cycle be affected in chloroplasts poisoned by dinitrophenol?

- A No change in rate as Calvin cycle occurs in the stroma and not at thylakoid membranes.
- B The rate of Calvin cycle will increase as pH in the stroma will decrease towards the optimum for enzymes involved in the cycle.
- C The rate of Calvin cycle will decrease with the accumulation of glycerate-3-phosphate.
- D The rate of Calvin cycle will decrease with the accumulation of glyceraldehyde-3-phosphate.

- 16 Isotopes of oxygen can be used to distinguish between oxygen absorbed by plants and oxygen evolved.

A mixture of oxygen isotopes $^{16}\text{O}_2$ and $^{18}\text{O}_2$ was supplied to a suspension of the unicellular alga *Chlorella* which had previously been exposed to $^{16}\text{O}_2$ only. During the following hour, changes in concentration of these gases in the suspension were measured in light and dark conditions.

The graph shows the results.



What caused the concentration of $^{16}\text{O}_2$ to rise in light?

- A H_2^{18}O was being photolysed more rapidly than H_2^{16}O .
 - B $^{16}\text{O}_2$ was being absorbed at different rates in light and dark.
 - C $^{16}\text{O}_2$ was being produced in photosynthesis but was not being absorbed in respiration.
 - D $^{16}\text{O}_2$ was being produced in photosynthesis faster than it was being absorbed in respiration.
- 17 Both glucose and appropriate enzymes are necessary for the process of glycolysis to begin.

Which additional compound must also be present?

- A Acetylcoenzyme A
- B ATP
- C Pyruvate
- D Reduced NAD

18 In which of the following setups would carbon dioxide be produced?

	I	II	III
A	Mitochondria + glucose	Cell lysate + pyruvate	Chloroplast + RuBP carboxylase
B	Mitochondria + starch + amylase	Cell lysate + RuBP carboxylase	Cell lysate + pyruvate
C	Cell lysate + glucose	Cell lysate + pyruvate	Mitochondria + pyruvate
D	Cell lysate + pyruvate	Chloroplast + RuBP carboxylase	Lysosome + starch

19 Cyanide is a poison that prevents the production of ATP.

What will happen if a resting axon is treated with such a poison?

- A The charge outside the axon membrane does not change.
- B The charge inside the axon membrane becomes more negative.
- C The charge outside the axon membrane becomes less positive.
- D The charge inside the axon membrane becomes less positive.

20 Molecular homology of haemoglobin is used to determine evolutionary relatedness in a range of mammals. Either nucleic acids or proteins can be sequenced. The sequence of nucleic acids and proteins are compared.

Which sequence will vary **LEAST** in related mammals?

- A Amino acid
- B DNA
- C Edited mRNA transcript
- D Primary mRNA transcript

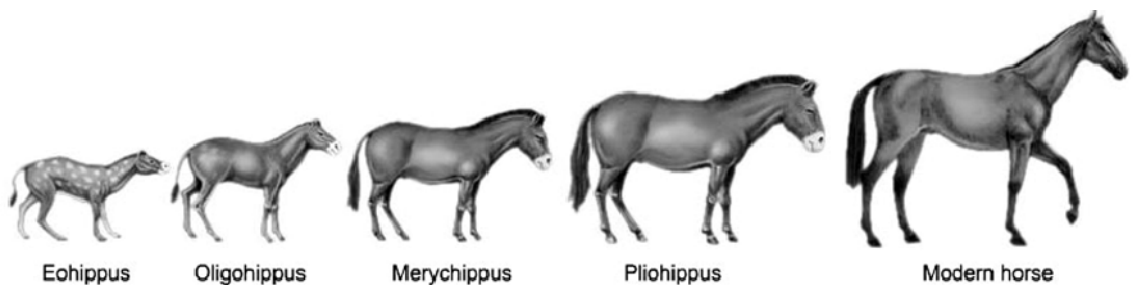
- 21 When organochlorine insecticides such as DDT were in widespread use, mosquitoes in malarial regions developed resistance more rapidly than houseflies in Britain.

What could account for the difference in the rates of the development of resistance?

- A More insecticides were used in Britain.
 - B More insecticides were used in malarial regions.
 - C Mosquitoes have fewer random mutations when exposed to insecticides.
 - D Mosquitoes have more random mutations when exposed to insecticides.
- 22 Fish in the Arctic and Antarctica have antifreeze proteins. But these two populations of fish diverged long before the antifreeze genes evolved. It was found that the genes that code for the antifreeze proteins in fish, at the Arctic and Antarctica, are quite different. This is evidence that separate and independent episodes of molecular evolution occurred, and obtained same functional result.

Which of the following processes is responsible for the antifreeze property in the two populations of the fish?

- A Reproductive isolation
 - B Divergent evolution
 - C Convergent evolution
 - D Geographical isolation
- 23 The diagram below suggests the evolution of horses beginning from the Eohippus 58 million years ago.

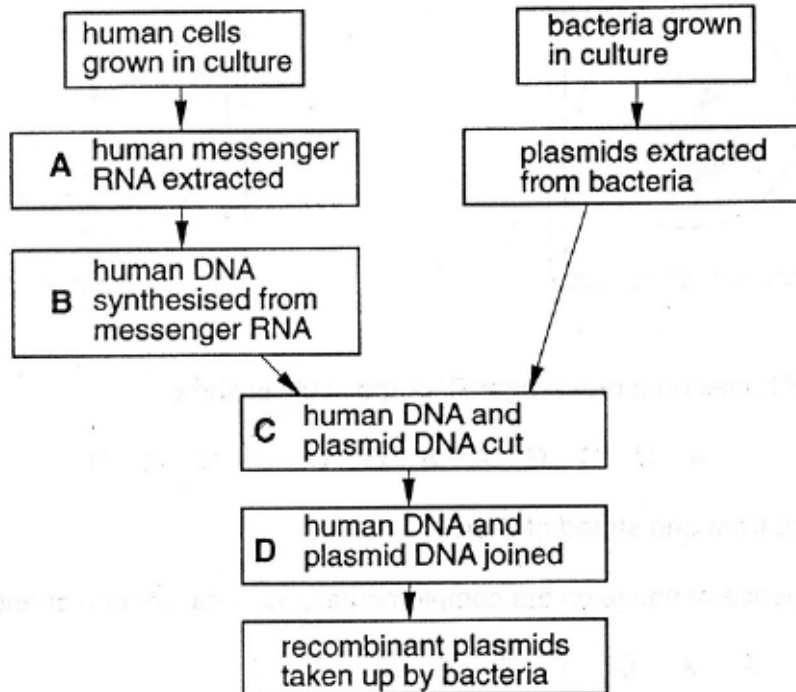


Fossil records suggest that the ancestor of the modern horse is believed to have had relatively short legs. According to Darwinian views, the evolution of long-legged horses is best explained by _____.

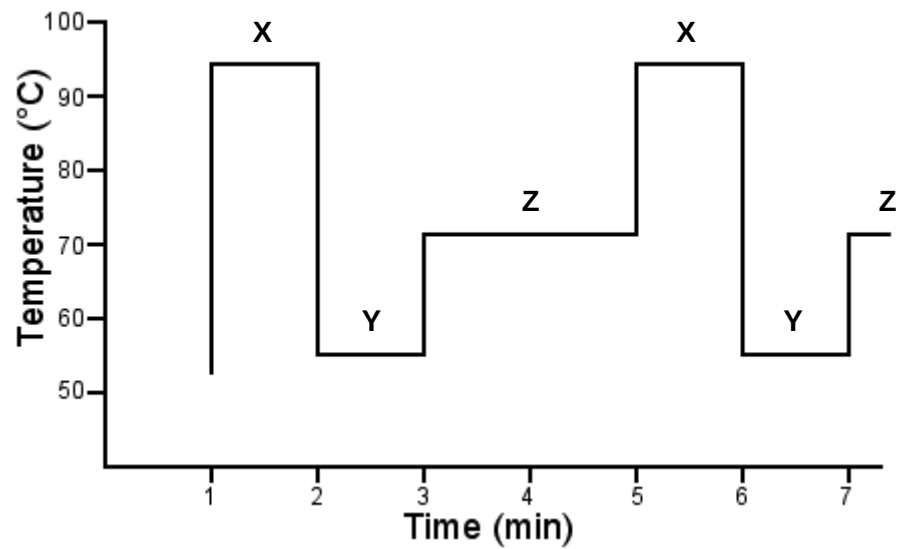
- A stabilising selection
- B directional selection
- C divergent evolution
- D descent by modification

- 24 The flow chart below shows some of the events in the production of a human hormone by genetic engineering.

At which stage in the process is a restriction enzyme used?



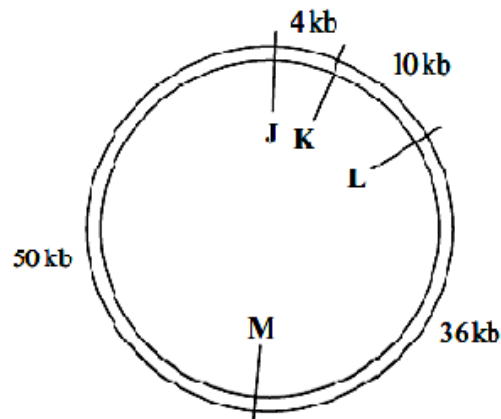
- 25 The diagram below shows the changes in temperature in a thermocycler over time during polymerase chain reaction.



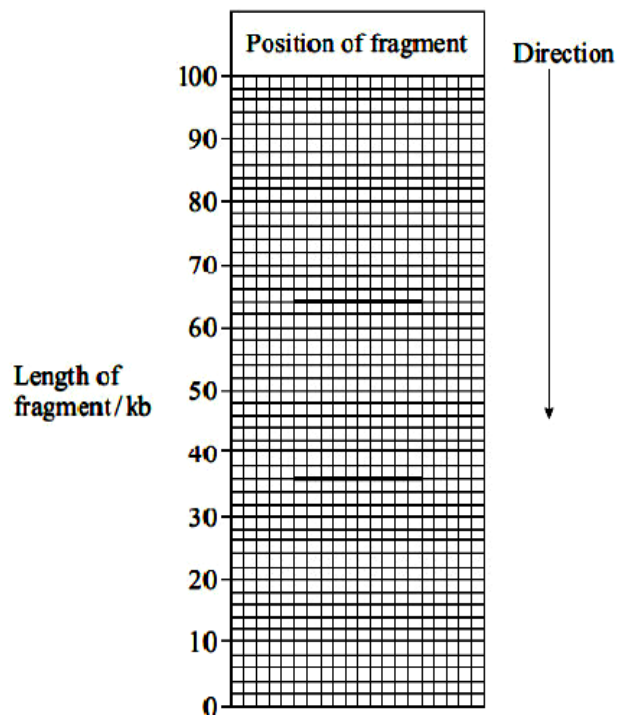
Which of the following statements is **TRUE**?

- A Elongation of new daughter strand occurs during Y.
- B Taq polymerase functions optimally at X.
- C DNA primers are anneal to the DNA template during Z.
- D Double stranded DNA template denatures into single strands during X.

- 26 The diagram below shows the position of four restriction sites J, K, L and M for four different restriction enzymes in a single plasmid. The distances between these sites are measured in kilobases of DNA.



The plasmid was cut using only two restriction enzymes. The resulting fragments were separated by gel electrophoresis. The positions of the fragments are shown in the chart below.

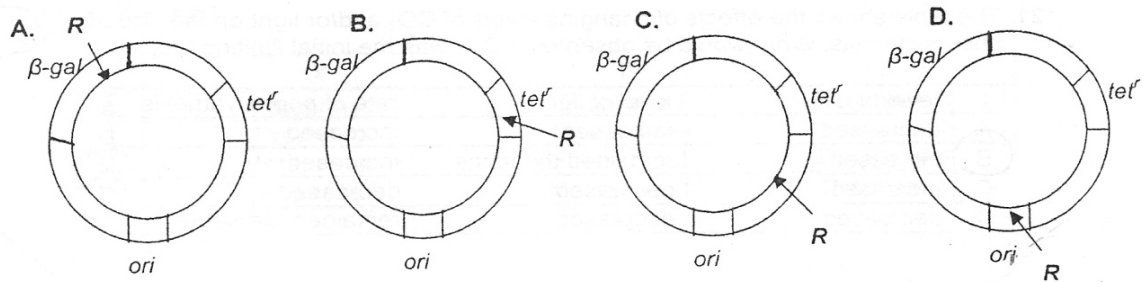


Which restriction sites were cut?

- A J and K
- B L and M
- C J and M
- D L and K

- 27 A synthetic plasmid having an origin of replication (*ori*), a restriction site (*R*), two genetic markers, β -galactosidase gene (*β -gal*) and tetracycline-resistance gene (*tet^r*), was used as a vector to transfer the human growth hormone gene into bacteria. The *β -gal* gene codes for an enzyme β -galactosidase, which breaks down a colourless compound called X-gal to a blue compound.

At which position should the restriction site (*R*) be located on the plasmid in order to identify the desired transformed bacteria on an agar medium containing both X-gal and tetracycline?



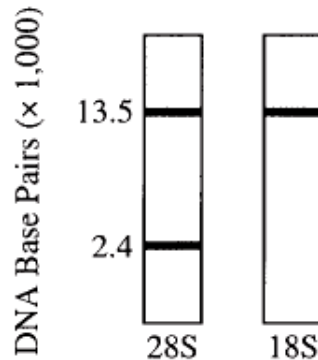
- 28 How is the polymerase chain reaction (PCR) similar to the replication of DNA?

- I DNA is heated to break hydrogen bonds
- II DNA unzips
- III free nucleotides are used
- IV DNA polymerase are required

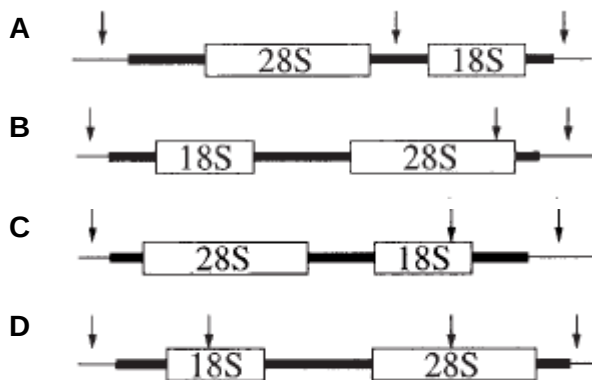
- A I only
- B II and IV only
- C I, II and III only
- D II, III and IV only

- 29 The autoradiograms obtained below (after electrophoresis and Southern Blotting) show human DNA digested with a specific restriction enzyme and probed with labeled rRNA.

On the left, the probe used was 28S rRNA. On the right, the probe used was 18S rRNA.



If the arrows in the following map show the location of the restriction sites of this restriction enzyme, which map **BEST** explains the results shown above?



- 30 Which of the following explains why insulin expressed in a prokaryotic host cell is not functional?

- I. Prokaryotic cells cannot replicate human DNA.
- II. Prokaryotic cells cannot splice mRNA.
- III. Prokaryotic cells do not possess sER.
- IV. Prokaryotic protein expression machinery stops at translation.

- A I only
- B II only
- C II and III
- D I, II, III and IV

END OF PAPER

2013 Y6 Preliminary Exam H1
MCQ Answer Scheme

1	B	16	D
2	C	17	B
3	B	18	C
4	B	19	A
5	D	20	A
6	D	21	B
7	A	22	C
8	C	23	D
9	D	24	C
10	B	25	D
11	A	26	B
12	B	27	A
13	A	28	D
14	B	29	B
15	C	30	B